CENG 252, Fall 2026, HW#1

Instructor: Zheng Chen (Email: zhc199@ucsd.edu)

Due - 8:59pm, Oct. 9, 2026

Problem 1. An exothermic reaction A → B + C was carried out adiabatically and the following data was recorded:

X_A	0	0.2	0.4	0.5	0.6	0.8	0.9
$-r_A$ (mol/L-min)	10	16.67	50	50	50	12.5	9.09
$1/-r_A$ (L-min/mol)	0.1	0.06	0.02	0.02	0.02	0.08	0.11

The entering molar flow rate of A was 300 mol/min, please find:
(a) What PFR volume is necessary to achieve 40% conversion?
(b) What conversion can be achieved if the PFR in part (a) is followed by a 2.4 L CSTR?
(*Note – "exothermic reaction": a reaction that generates heat; "adiabatic": no heat exchange)*

Problem 2. The reaction: A → B is to be carried out isothermally in a continuous-flow reactor. Calculate both the CSTR and PFR reactor volumes necessary to consume 99% of A (i.e., C_A = 0.01 C_{A0}) when the entering molar flow rate is N_{A0} = 5 mol/h, assuming the reaction rate $-r_A$ is:

(a) $-r_A = k$, with $k = 0.05$ mol h^{-1} dm^{-3};

(b) $-r_A = kC_A$, with $k = 0.0001 s^{-1}$;

(c) $-r_A = kC_A^2$, with $k = 3$ dm^3 mol^{-1} h^{-1};

The entering volumetric flow rate is $v_0 = 10$ dm^3/h. (Note: $N_A = C_A v$. For a constant volumetric flow rate $v = v_0$, then $N_A = C_A v_0$. Also, $C_{A0} = N_{A0}/v_0$)

(d) Repeat (a) and (b) to calculate the time necessary to consume 99.9% of species A in a 1000 dm^3 constant volume batch reactor with C_{A0} = 0.5 mol dm^{-3}.

Problem 3. A gas phase elementary reaction takes place in PFR as the following:

$$2A \rightarrow B$$

The initial temperature is 500 K and the pressure is 8.2 atm. Only A is fed into the reactor with a volumetric flow rate of 25 dm^3/s and concentration of 0.2 mol/dm^3. The reaction rate constant is found to be k = 10 dm^3/(mol·s). What will be the size of the reactor for achieving $X = 90\%$? What is τ *(time*? assuming the pressure and temperature of the reactor maintains the same during reaction.

Problem 4: The elementary gas-phase reaction

$$(CH_3)_3COOC(CH_3)_3 \rightarrow C_2H_6 + 2CH_3COCH_3$$

is carried out isothermally at 400K in a flow reactor with no pressure drop. The specific reaction rate at 50°C is 10^{-4} min^{-1} (from pericosity data) and the activation energy is 85 kJ/mol. Pure di-tert-butyl peroxide enters the reactor at 10 atm and 127°C and a molar flow rate of 2.5 mol/min. Calculate the reactor volume and space-time to achieve 90% conversion in

(a) A CSTR
(b) A PFR
(c) Assume that the reaction is reversible with $K_c = 0.025$ mol^2/dm^6, and calculate the equilibrium conversion; then redo (a) and (b) for achieving 90% of equilibrium conversion.

Extra problems for Non-Chemical Engineers:
Go through examples in Folger chapters for more practice.